View Reviews

Paper ID
2366

Paper Title
CLIP-CAT: Vision-Language Fusion Framework for Ordinal Cataract Grading on Few-Shot Data

Reviewer #1

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html
MIC

4. How would you describe the paper?
Methodological contribution

5. Please describe the main contribution of the paper.
The paper proposes a vision-language framework, CLIP-CAT, for cataract grading that integrates global semantic features from a pretrained CLIP encoder with locally extracted fine-grained features via an adaptive patching mechanism, and introduces a rank-aware loss to model the ordinal nature of disease severity. By combining these components, the method aims to improve grading performance on small and imbalanced clinical datasets.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.
1）Clinically motivated ordinal modeling: The paper explicitly incorporates the ordinal nature of cataract grading via a rank-aware loss, which is well aligned with clinical practice where the severity levels follow a natural progression and large grading errors are more critical than minor ones.
2）Reasonable integration of global and local feature learning: The proposed dual-branch design combines global representations from a pretrained CLIP encoder with locally extracted fine-grained features, which is intuitively suitable for capturing both overall lens structure and subtle opacity patterns under limited data conditions.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of

clinical feasibility, or application is not novel, then you must provide specific references to prior work.

1. Background and Problem Setting

• Weak clinical motivation: The practical necessity of cataract grading based on slit-lamp images is not sufficiently justified. The paper lacks discussion on real deployment scenarios and does not clarify its advantages over other commonly used modalities (e.g., OCTA or fundus imaging).

• Misleading “few-shot” claim: The paper does not follow a standard few-shot learning protocol (e.g., N-way K-shot setting), and is essentially trained on a small-scale dataset, making the claim potentially misleading.

• Incomplete related work: The review of existing cataract grading and ordinal modeling methods is insufficient, leading to unclear positioning of the proposed approach within the literature.

2. Methodological Limitations

• Limited methodological novelty: The overall framework lacks a coherent design principle, and the components (global branch, local branch, text alignment, and ordinal loss) appear loosely connected, resembling an engineering combination rather than a unified modeling framework.

• Questionable contribution of vision-language modeling: The prompt design is tightly coupled with class labels and essentially functions as a textual reformulation of labels. There is insufficient evidence that the model achieves genuine semantic understanding.

• Lack of medical justification for local modeling: The entropy-based patch selection strategy lacks clinical grounding. Cataracts primarily manifest as opacity changes within the lens region, rather than increased texture complexity. High-entropy regions may arise from illumination variations, reflections, or noise, and do not necessarily correspond to pathological areas.

• Unclear source of performance gains: The observed improvements may largely stem from the use of a stronger pretrained backbone and ordinal loss, rather than the claimed semantic reasoning mechanism.

3. Limited data scale with risk of overfitting: The dataset contains only approximately 2,000 images from a single center. Under such conditions, the model may overfit to fixed prompts and label structures, rather than learning robust representations.

4. Experimental Design and Evaluation

• Insufficient comparison: The set of compared methods is limited and does not include recent or task-specific approaches, making it difficult to convincingly demonstrate superiority.

• Incomplete analysis: The paper lacks per-grade performance evaluation and confusion matrix analysis, making it unclear how the model performs across different severity levels.

• Limited validation scope: Experiments are conducted on a single dataset only, with no broader validation to support the robustness of the conclusions.

5. Presentation and Structural Issues

• Inconsistency between figures and text: Fig. 1 includes components (e.g., multiple

types of text inputs, Cross-Model Fusion, and report generation) that are not described in the main text, leading to inconsistencies in the method description.
• Unclear figure representation: Fig. 2 contains multiple arrows and connections that are not clearly explained and do not align well with the textual description, making the figure difficult to interpret.
• Lack of writing rigor: Some sections exhibit mismatches between subsection titles and content, along with promotional or imprecise descriptions, reducing the overall technical clarity and rigor of the paper.

8. Please rate the clarity and organization of the paper.
Poor

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.
The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/
No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.
1. Strong Reject — must be rejected due to major flaws

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?
The paper is not recommended for acceptance due to limited methodological novelty, insufficient justification of key design choices, small-scale data with potential overfitting, and incomplete experimental validation, which together weaken the overall contribution and credibility of the work.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?
Very confident (4)

Reviewer #2

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html

MIC

4. How would you describe the paper?

Methodological contribution

5. Please describe the main contribution of the paper.

1. This paper proposes CLIP-CAT, a vision language modal that leverages medical text prompts to alleviate data scarcity and improve grading performance.
2. It combines language-guided global semantic representations and localized fine-grained feature learning in a unified framework and introduces Adaptive Patching for enhanced local representation.
3. It designs a Bidirectional Rank-Aware Loss that encourages unimodal probability distributions.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

1. The paper is in good writing and easy to follow.
2. The authors propose a practical approach to solve a realistic clinical problem.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

1. The authors don't present clear formula for the Bidirectional Rank Loss, which is a key part in this paper. The review cannot evaluate this paper without such important detail.
2. The approach is lack of novelty. The authors introduce CLIP, global-local attention in their network, which are very popular nowadays.
3. The evaluation of this approach is not enough. The authors only compare their method with two CNN models and one ViT model, which are not the state-of-the-art methods.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/

No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.
2. Reject — should be rejected, independent of rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?
As I mentioned in Weaknesses, the authors have to address several concerns about their model evaluation, method novelty, and approach details.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?
Very confident (4)

Reviewer #3

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html
CAI

4. How would you describe the paper?
Methodological contribution

5. Please describe the main contribution of the paper.
The manuscript proposes CLIP-CAT, a vision–language fusion framework for ordinal cataract grading in few-shot settings, combining language-guided global semantic representations with localized fine-grained feature learning.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.
The key strength of the work is the incorporation of multi-granularity learning and a bidirectional rank-aware constraint in the classification loss, which is well aligned with the ordinal nature of cataract grading.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.
However, the manuscript has several limitations:
- The level of novelty appears limited, as similar vision–language and ranking-based approaches have been explored in prior work with various tasks.

- The paper would benefit from substantial revision, as it currently reads more like a technical report than a well-structured academic manuscript.
- Important training details for the encoders are missing, and the lack of publicly available code raises concerns regarding reproducibility.
- The proposed approach has not been validated on external datasets, which limits confidence in its generalizability.
- Providing confusion matrices would improve the interpretability of model performance across classes.
- Overall, the technical contribution is not sufficiently strong or clearly articulated.

8. Please rate the clarity and organization of the paper.
Poor

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.
The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/
No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.
2. Reject — should be rejected, independent of rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?
The paper addresses an important problem and proposes a vision–language framework with multi-granularity and rank-aware learning. However, the overall contribution is limited by a lack of novelty, insufficient methodological details, and concerns regarding reproducibility. Additionally, the absence of external validation and limited performance analysis reduce confidence in the generalizability and impact of the proposed approach.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?
Very confident (4)